Geophysical Journal International

Geophys. J. Int. (2023) **234,** 1640–1654
Advance Access publication 2023 April 18
GJI Geomagnetism and Electromagnetism

https://doi.org/10.1093/gji/ggad166

Palaeosecular variation in Northern Patagonia recorded by 0–5 Ma Caviahue–Copahue lava flows

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Accepted 2023 April 13. Received 2023 April 5; in original form 2022 December 28

SUMMARY

Palaeosecular variation (PSV) determinations and studies of the geometry of the Earth's main magnetic field provide important information about the field evolution, and to constrain numerical geodynamo models. Palaeomagnetic directional data from lava flows over the last few million years is of particular interest because the regional and global tectonic effects are minimal. However, the distribution of this type of palaeomagnetic data is uneven where the Southern Hemisphere is the destitute side. Therefore, the better knowledge of the geomagnetic field behaviour depends on the increased availability of high-quality data, especially in the Southern Hemisphere. A PSV and the time-averaged field (TAF) study was then performed in 0–5 Ma lava flows from the Caviahue–Copahue Volcanic Complex located in Northern Patagonia, Argentina (37°0′S, 71°10′W). The magnetic mineralogy of lava flows was investigated through thermomagnetic susceptibility curves, isothermal remanent magnetization (IRM) acquisition curves, hysteresis loops and first-order reversal curves (FORCs). Samples are essentially comprised of titanomagnetite with different Ti contents and magnetic domain structures typical of vortex state particles. A total of 50 volcanic sites were sampled, which provided 42 reliable palaeomagnetic site-mean directions after alternating field and thermal demagnetization. From these 42 sites, 36 are of normal and 6 are of reversed polarity. The mean direction from normal (reversed) sites is $D = 356.2°$, $I = -50.1°$, $\alpha_{95} = 4.0°$ and $N = 36$ ($D = 176.5°$, $I = 59.5°$, $\alpha_{95} = 14.1°$ and $N = 6$). Using only site-level data with the precision parameter $k \geq 100$, we obtain 26 palaeomagnetic sites for PSV and TAF investigations in the study region. The filtered data set has a mean direction ($D = 354.4°$, $I = -53.2°$ and $\alpha_{95} = 5.1°$) close to the expected direction for a geocentric axial dipole (GAD) field ($I_{GAD} = -57.3°$). The palaeopole (Plat = 84.4°, Plon = 229.1° and $A_{95} = 5.7°$) coincides with the Earth's spin axis within the 95 per cent confidence interval. Virtual geomagnetic pole scatter ($S_B = 15.8^{18.9°}_{11.8}$) and the inclination anomaly ($\Delta I = 4.1^{9.2°}_{-1.0}$) are both consistent at the 95 per cent confidence level with recent PSV and TAF models, respectively. Our results support the presence of small non-dipole field contributions (<3 per cent) superimposed on the GAD term, as reported by South American studies at mid southern latitudes.

Key words: Palaeomagnetic secular variation; Palaeomagnetism; Rock and mineral magnetism.

1 INTRODUCTION

An accurate diagnosis of the time-averaged palaeomagnetic field (TAF) and its secular variation is a key to improve our knowledge of the Earth's magnetic field structure. Over a long time interval of the order of 10^5–10^6 yr, TAF geometry can be described as a first approximation by a geocentric axial dipole (GAD). This hypothesis is accepted by the palaeomagnetic community with applications for plate tectonic reconstructions over geological timescales (McElhinny 2007; Tauxe 2010). However, for the past 10 Myr, recent

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studies of palaeosecular variation (PSV) suggest that the palaeomagnetic field has small non-GAD contributions, mainly due to the presence of axial quadrupolar and octupolar components (e.g. Johnson *et al.* 2008; Cromwell *et al.* 2018; Bono *et al.* 2020; de Oliveira *et al.* 2021). On these timescales, information about the TAF structure can be obtained from palaeomagnetic records of volcanic and sedimentary rocks. Particularly, lava flows are commonly used for statistical evaluations of the palaeomagnetic field because they provide geologically instantaneous records in contrast to the smoothed recording in sedimentary materials (Aubert *et al.* 2010; Hulot *et al.* 2010).

The 0–10 Ma interval is of particular interest because the regional and global tectonic effects are small and because of the large availability of palaeomagnetic data compared to older periods (Johnson & McFadden 2015). Palaeodirectional data recorded by lavas flows over the past 5 Myr were first compilated by McElhinny & Merrill (1975) and then updated by many PSV studies (e.g. Quidelleur *et al.* 1994; Johnson & Constable 1996; McElhinny & McFadden 1997; Johnson *et al.* 2008). These databases differ by inclusion of new data and the selection criteria used, following the evolution of modern palaeomagnetic laboratory methods. The current data set was presented by de Oliveira *et al.* (2021) that includes palaeodirectional data of volcanic rocks from selected palaeomagnetic studies for the past 10 Myr. This data set provides improvements in the spatio-temporal coverage of high-quality data relative to the previous PSV10 database (Cromwell *et al.* 2018), which was used to construct new PSV and TAF models.

Statistical investigations of PSV behaviour provide important information about the palaeomagnetic field evolution, and to constrain numerical geodynamo models (e.g. Christensen & Olson 2003; Coe & Glatzmaier 2006; Davies *et al.* 2008; Christensen *et al.* 2010; Sprain *et al.* 2019; Biggin *et al.* 2020). PSV is commonly investigated by means of the angular dispersion of virtual geomagnetic poles (VGPs) as a function of latitude (Johnson & McFadden 2015). Model G (McFadden *et al.* 1988) has been widely used to assess latitudinal variation of VGP dispersion over Phanerozoic times (e.g. Biggin *et al.* 2008; Veikkolainen & Pesonen 2014; de Oliveira *et al.* 2018; Doubrovine *et al.* 2019; Franco *et al.* 2019). An inverse relationship between VGP dispersion pattern and average geomagnetic reversal frequency was suggested by some studies (McFadden *et al.* 1991; Biggin *et al.* 2008). However, recent studies (Doubrovine *et al.* 2019; Handford *et al.* 2021) disputed this correspondence from new results of Model G parameters for Permian–Triassic transition (200–264 Ma), and similarity observed in VGP dispersion curves between periods with no reversals [Cretaceous Normal Superchron (84–126 Ma) and Permian–Carboniferous Reversed Superchron (262–318 Ma)] and 0–10 Ma interval. The latter characterized by a high reversal rate ($\sim$ 4.8 Myr^{-1}). Furthermore, Cromwell *et al.* (2018) and de Oliveira *et al.* (2021) suggest an equatorial PSV asymmetry, with higher VGP dispersion in the Southern Hemisphere than the Northern Hemisphere for the past 10 Myr. It is worth mentioning that Linder & Gilder (2012) suggested that latitudinal dependence of VGP dispersion might be a mathematical artefact, as a result of the transformation of directions to VGPs and assuming a latitude-invariant behaviour of direction scatter. Nonetheless, some studies (e.g. Veikkolainen & Pesonen 2014; Doubrovine *et al.* 2019) do not support this hypothesis.

Another important aspect concerns whether the palaeofield structure is essentially dipolar for the past few million years, as described by GAD hypothesis (Merrill & McFadden 2003). It is possible to investigate the TAF structure from the latitudinal behaviour of inclination anomaly (ΔI) data (McElhinny & McFadden 2000). A geomagnetic field model fitted to the ΔI data allows the determining the zonal quadrupole and octupole terms relative to the axial dipole term, based on spherical harmonic analysis (McElhinny *et al.* 1996). Some studies reveal the presence of non-dipolar contributions less than 5 per cent of the axial dipole for the 0–5 Ma interval (e.g. Merrill & McElhinny 1977; Schneider & Kent 1990; McElhinny *et al.* 1996; Johnson *et al.* 2008). Johnson *et al.* (2008) suggest contributions of 2–4 per cent for the quadrupole component and 1–5 per cent for the octupolar component. de Oliveira *et al.* (2021) proposed a zonal TAF model for the 0–10 Ma interval, with contributions of 3.2 and 1.2 per cent for the zonal quadrupole and octupole terms, respectively. A suggested approach for evaluating the spatial features (latitudinal and longitudinal variations) of the palaeomagnetic field includes non-zonal TAF contributions (e.g. Johnson & Constable 1995, 1997; Hatakeyama & Kono 2002; Cromwell *et al.* 2018). However, details of the palaeofield morphology are affected by the uneven spatial and temporal distributions of palaeomagnetic data. The scarcity of palaeomagnetic data is observed at low and high latitudes in both hemispheres, especially in the Southern Hemisphere that contributes only 35 per cent of the 0–10 Ma database (Cromwell *et al.* 2018; de Oliveira *et al.* 2021). Therefore, new palaeodirectional records are crucial for a better understanding the spatial structure and variability of the palaeomagnetic field.

This study presents new high-quality palaeomagnetic data from Pliocene–Holocene lava flows located in the Caviahue–Copahue Volcanic Field, northern Patagonia (Argentina). Moreover, we offer new VGP scatter and inclination anomaly estimates and compare to the South American studies for the past 10 Myr, in order to verify the compatibility of PSV and TAF models.

2 GEOLOGICAL SETTING

The Caviahue–Copahue Volcanic Complex (CCVC) is located in the Andean Southern Volcanic Zone (SVZ) close to the Chile–Argentina border (37°50′S, 71°10′W, Fig. 1a). In this area, volcanism resulted from oblique subduction of the Nazca Plate under the South American Plate (Stern 2004). The CCVC consists of Pliocene–Holocene effusive and explosive volcanic deposits mainly basaltic andesite and rhyolite compositions (Mazzoni & Licitra 2000; Varekamp *et al.* 2006). A volcanic tectonic depression, designated as Caviahue Caldera, covers an area of about 300 km^2 and 500 km deep, which was developed during the Upper Pliocene (2 Ma; Folguera & Ramos 2000). It is composed of abundant pyroclastic deposits found both inside and outside of the caldera (Linares *et al.* 1999; Folguera & Ramos 2000; Haag *et al.* 2021). Located at the western margin of the Caviahue Caldera, the Copahue Volcano comprises the active volcanic center of the study area (Fig. 1b), and consists of lava flows with ages $<$1.2 Ma (Linares *et al.* 1999). In this study, we focus on the effusive lavas from four geological units of the CCVC: Cola de Zorro Formation, Las Mellizas Formation (Lower and Upper lavas), Trolope Lavas and Copahue Volcano. An overview of each geological unit is presented as follows.

2.1 Cola de Zorro Formation

The eruptive history of the CCVC started with effusive and pyroclastic eruptions formed under an extensional tectonic regime (Folguera *et al.* 2003), whose deposits are associated with the Cola de Zorro Formation (Figs 2 and 3g). This unit is characterized by subhorizontal basalt and andesite lava flows, generally interbedded with

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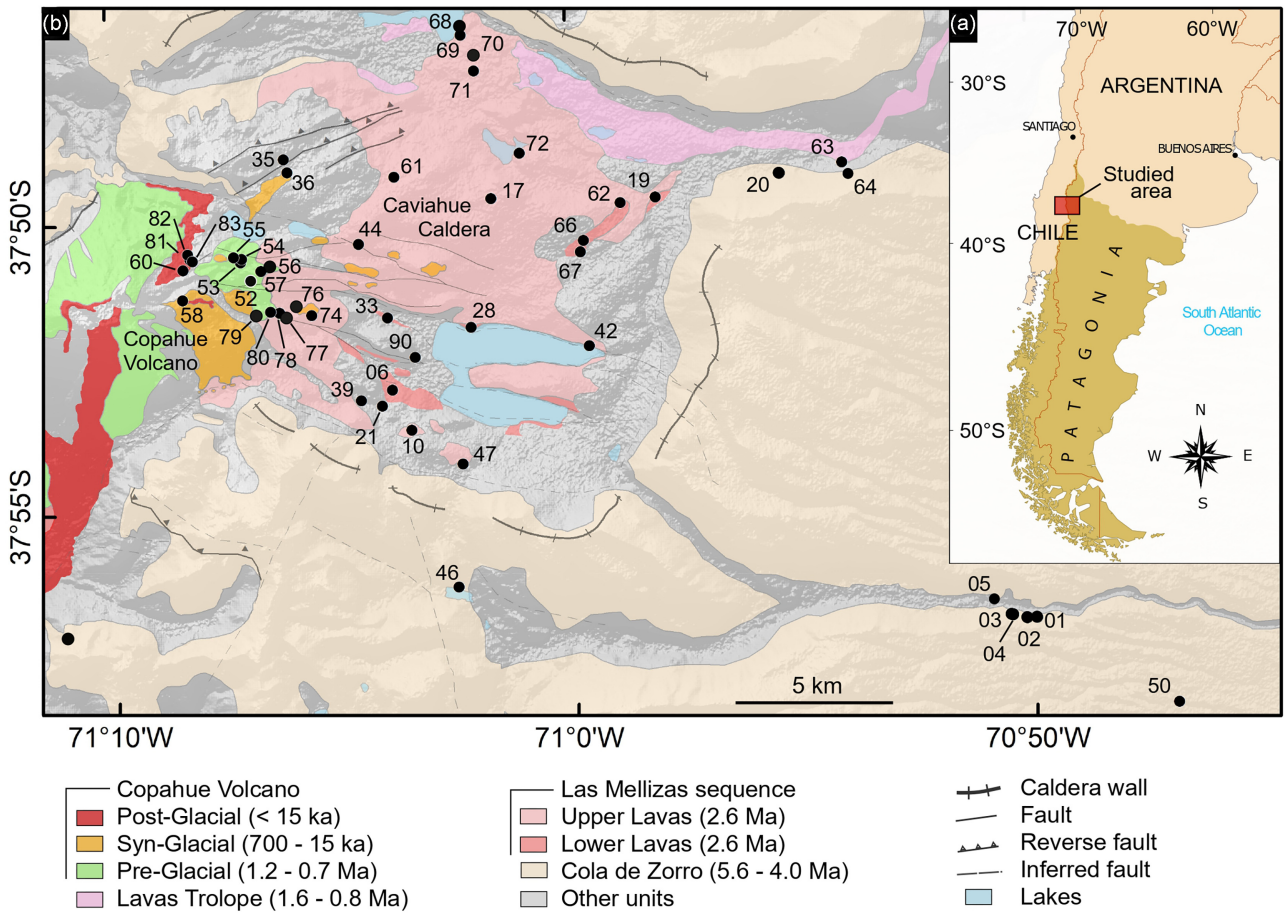


Figure 1. (a) Location map of South America showing the studied area. (b) Digital elevation model from modified geological map of CCVC (Melnick *et al.* 2006). The points indicate sampled sites.

non-welded ignimbrites of andesitic composition (Melnick *et al.* 2006). Its volcanic deposits range in age from 5.67 to 4.00 Ma (see Table S1, Supporting Information), and covers a large area situated to the west of the CCVC (Muñoz & Stern 1988; Linares *et al.* 1999). In this geological formation 72 rock cores were sampled.

2.2 Las Mellizas

The Las Mellizas Volcanic Sequence comprises two effusive events (Lower and Upper Lavas) intercalated by an explosive one (Las Mellizas Ignimbrites). The Lower Lavas are represented by columnar-jointed lavas of basaltic to andesitic composition, locally interbedded with sedimentary deposits (Figs 2 and 3d). This unit is emplaced above the Cola de Zorro Formation and is widespread inside the Caviahue Caldera (Fig. 1b). In contrast, the Upper Lavas overlap the Las Mellizas Ignimbrites and are composed of broadly homogeneous andesitic lava flows with subhorizontal foliation (Figs 2 and 3e). The entire Las Mellizas sequence is constrained by K–Ar ages around 2.6 Ma (Table S1, Supporting Information; Linares *et al.* 1999). A total of 184 rock core samples were taken in this geological sequence.

2.3 Trolope Lavas

Situated in the northeast portion of CCVC, the Trolope Lavas comprise a 200 m thick effusive sequence of nearly homogeneous trachyandesite composition (Figs 2 and 3f). These deposits were emplaced from 1.63 to 0.82 Ma (Table S1, Supporting Information; Linares *et al.* 1999) and 12 rock cores were extracted.

2.4 Copahue Volcano

The Copahue Volcano lava flows are associated with successive fissure and volcanic eruptions over the last 1.2 Ma (Table S1, Supporting Information; Muñoz & Stern 1988; Linares *et al.* 1999). Several stratigraphic schemes have been proposed for this unit (e.g. Melnick *et al.* 2006; Báez *et al.* 2020). Melnick *et al.* (2006) suggested three eruptive stages (Figs 2 and 3a–c). (1) The Pre-glacial stage (1.2–0.7 Ma), characterized by voluminous basaltic to andesitic lava flows that formed the main volcanic edifice. (2) The Syn-glacial stage (700–15 ka) is composed of andesitic to dacitic lava flows and supporting evidence for magma–water interactions, including pillow-lavas. (3) The Post-glacial stage (<15 ka) comprises basaltic–andesitic compositions and no evidence of glacial erosion. Recently, however, Báez *et al.* (2020) proposed a model of glaciovolcanic evolution with at least nine flow units organized into two sequences: the Ancient and Young sequences, separated by a major erosive episode. Furthermore, the Copahue Volcano also includes historical eruptions during 1992, 1995, 2000 and 2012 (Caselli *et al.* 2016), which involves ash, bombs and lithic fragments, besides rare lahar deposits (Melnick *et al.* 2006). For this formation, 136 rock cores were sampled.

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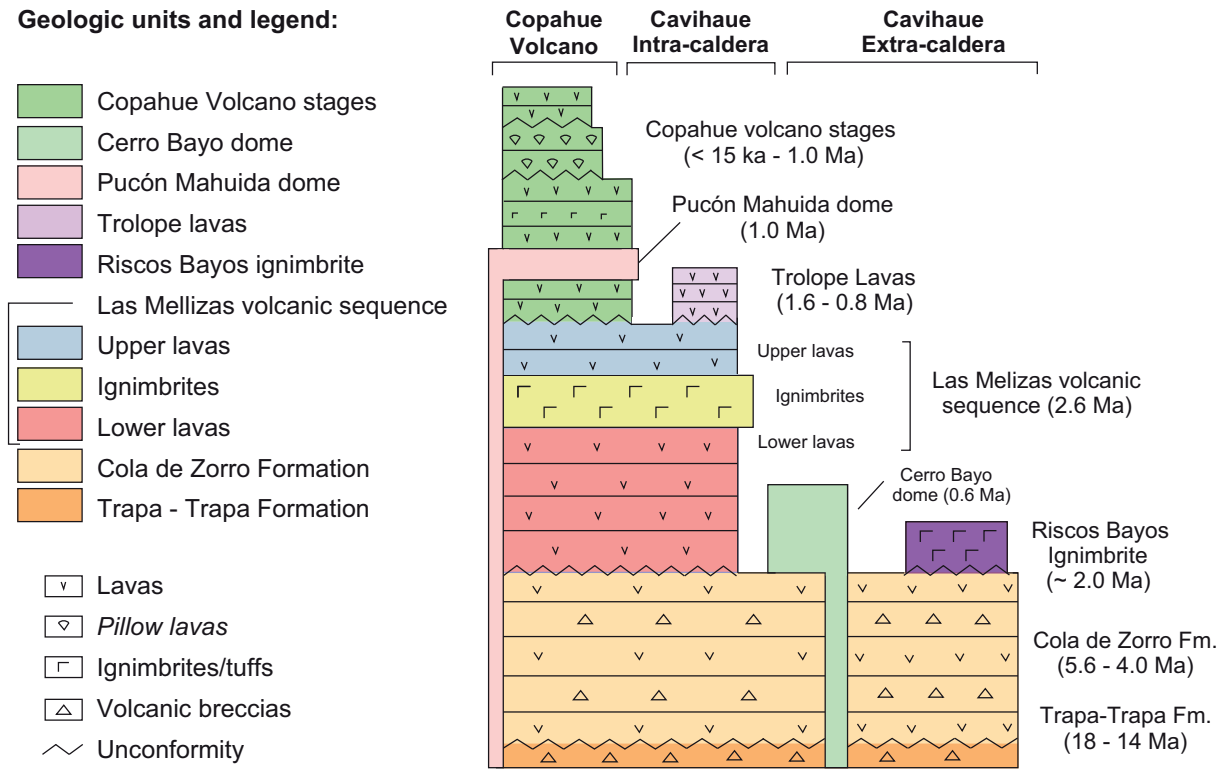


Figure 2. Stratigraphic chart of the Caviahue–Copahue Volcanic Field. Modified of Melnick *et al.* (2006).

3 SAMPLING

We sampled 50 independent sites during two field sessions in 2016 April and 2017 February (Fig. 1b). Palaeomagnetic samples were collected from four geological units within the study region (see Section 2). Recent glaciations offer a wealth exposure of effusive volcanic deposits, allowing the sampling of extensively unaltered bedrock outcrops. These effusive flows comprise the last 5 Myr and present variable compositions, ranging from basaltic to andesitic lava flows (Varekamp *et al.* 2006; Folguera *et al.* 2016). The ages assigned to the sampled sites were based on geochronological and volcanostratigraphic studies (Linares *et al.* 1999; Melnick *et al.* 2006), as reported in Table 1. Using a portable gasoline powered rock drill, at least 10 cores (2.5 cm in diameter and ∼12 cm in length) were taken from each site and oriented with a magnetic compass and when possible using a sun compass. No significant difference was observed between sun and magnetic compass readings.

4 METHODS

4.1 Rock magnetism measurements

All palaeomagnetic and rock magnetism measurements were carried out in the Palaeomagnetism Laboratory of the University of São Paulo (USPMag). From 15 representative sites, magnetic mineralogy experiments were done to assess the magnetic carriers and magnetic domain structures. The temperature variation of magnetic susceptibility (χ–T diagrams) was measured on one sample per site using a KLY4 Kappabridge coupled with a CS3 furnace (AGICO). Heating–cooling cycles were performed in air from room temperature up to 600 °C. The isothermal remanent magnetization (IRM) acquisition curves, hysteresis loops, and first-order reversal curves (FORC) diagrams were performed on one small chip sample from selected sites using a Princeton Measurements Corporation vibrating sample magnetometer (VSM) MicroMag 3900, with applying fields of up to 1 T at room temperature. FORC diagrams were obtained after 300 reversal curves with an averaging time of 200 ms. A smoothing factor of 4 was applied to all FORC measurements using the FORCinel software (Harrison & Feinberg 2008). Here, we provided new data from Copahue formation and rock magnetism data published by Moncinhatto *et al.* (2020).

5 PALAEODIRECTION EXPERIMENTS AND DATA ANALYSIS

Palaeodirection experiments were performed for the 50 sampled sites using a 2G Enterprises DC SQUID magnetometer housed in a magnetically shielded room with a background field less than 500 nT. About 5–10 specimens (1.1 cm in length) per site were subjected to stepwise alternating field demagnetization (AFD) in 15 steps up to 100 mT. In addition, three samples per site were submitted to stepwise thermal demagnetization (THD) in 10 steps up to 400 °C, followed by 50 °C steps until heating to 600 °C or complete removal of the natural remanent magnetization (NRM).

Demagnetization data were analysed using the Demag-GUI program that belongs to the PmagPy software package (Tauxe *et al.* 2016). Characteristic remanent magnetization (ChRM) directions at the specimen level were determined using principal component analysis (PCA; Kirschvink 1980) from Zijderveld diagrams (Zijderveld 1967). To estimate the best-fitting line of the ChRM components, we considered at least five consecutive demagnetization steps converging to origin of the orthogonal projection, with a maximum angular deviation (MAD; Kirschvink 1980) and Deviation Angle (DANG; Tauxe & Staudigel 2004) $\leq 5°$. MAD is a

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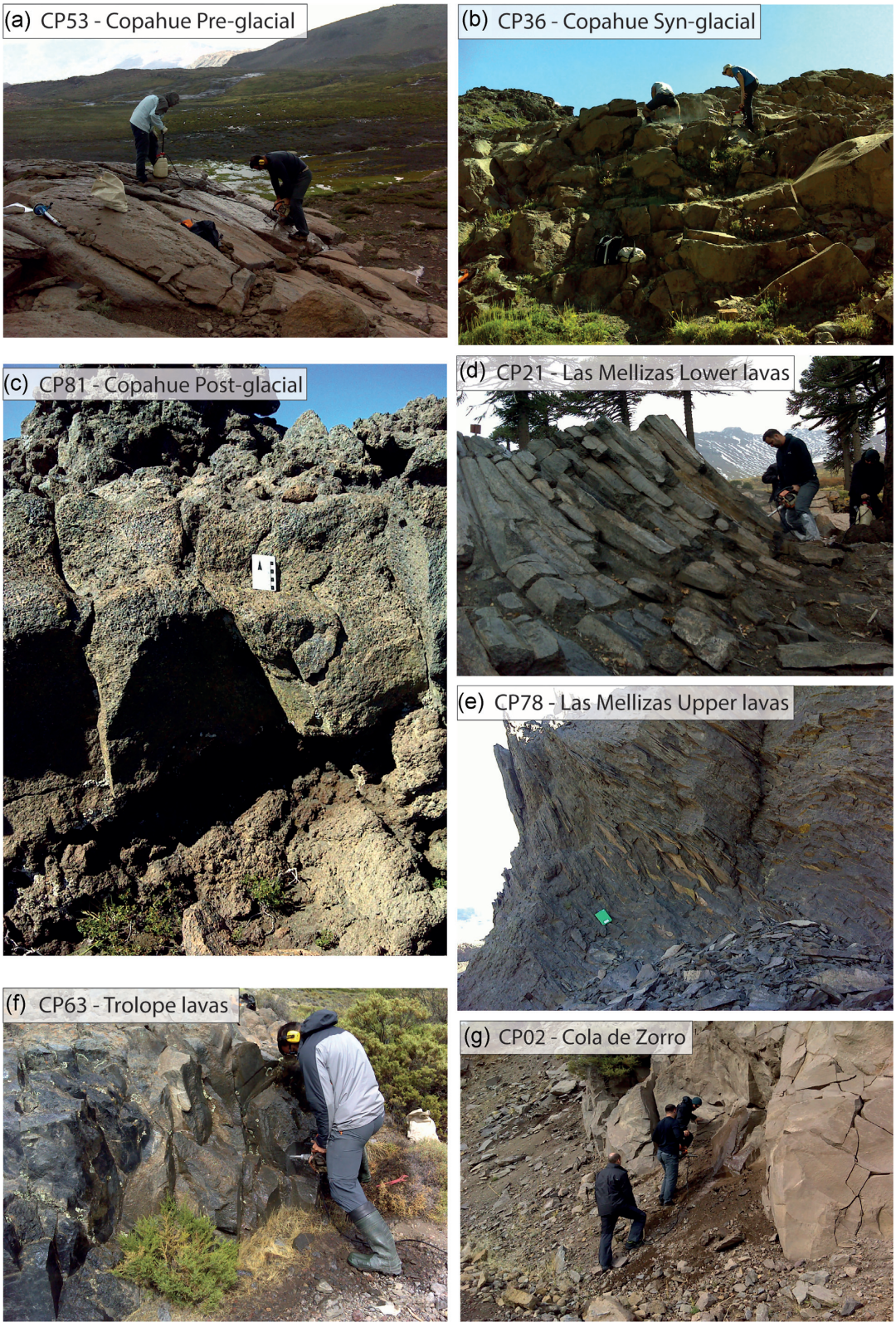


Figure 3. Images of CCVC units from different sampling sites. (a) Copahue Pre-glacial, (b) Copahue Syn-glacial, (c) Copahue Post-glacial, (d) Las Mellizas Lower lavas, (e) Las Mellizas Upper lavas, (f) Trolope lavas and (g) Cola de Zorro. In (c) and (e), the card and notebook serve as scale in the field. CCVC = Caviahue–Copahue Volcanic Complex.

measure of angular uncertainty of the best-fitting line and plane, while DANG corresponds to the angle of the ChRM direction in relation to the origin. Great circle fits (McFadden & McElhinny 1988) were carried out when the secondary component overlaps the primary component of magnetization. As adopted in the best-fitting lines, great circles with a MAD $\leq 5^{\circ}$ were deemed acceptable. Site-mean directions were calculated using Fisher statistics (Fisher 1953).

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Table 1. Palaeodirectional results from Caviahue–Copahue Volcanic Field.

Site	Slat (°N)	Slon (°E)	*n*	Dec (°)	Inc (°)	a_{95} (°)	*k*	Plat (°N)	Plon (°E)	P	Age (Ma)	Ref
CP01	−37.95148	−70.8328	8	191.7	69.2	3.1	145	73.1	264.2	R	4.0–5.6	1
CP02	−37.95150	−70.8363	7	196.9	70.7	2.6	252	69.5	260.8	R	4.0–5.6	1
CP03	−37.95058	−70.8416	6	193.6	52.8	4.5	104	78.0	180.7	R	4.0–5.6	1
CP04	−37.95049	−70.8420	6	191.6	60.5	6.9	45	80.4	224.3	R	4.0–5.6	1
CP05	−37.95054	−70.8418	5	147.0	37.4	9.2	180	56.8	40.6	R	4.0–5.6	1
CP06	−37.88233	−71.0650	9	344.8	− 55.5	1.6	552	77.7	202.9	N	2.5–2.8	1
CP10	−37.89397	−71.0584	9	356.5	− 42.8	4.3	95	76.6	275.1	N	2.5–2.8	1
CP17	−37.82789	−71.0279	9	9.6	− 50.3	1.7	256	79.6	341.2	N	2.5–2.8	1
CP19	−37.82845	−70.9683	8	25.4	− 35.6	1.3	372	61.4	346.7	N	2.5–2.8	1
CP20	−37.82228	−70.9233	7	163.8	56.2	2.5	209	77.1	18.8	R	4.0–5.6	1
CP21	−37.88693	−71.0689	8	351.4	− 47.7	6.2	69	78.4	248.2	N	2.5–2.8	1
CP28	−37.86481	−71.0361	7	339.5	− 48.7	4.5	84	71.1	218.7	N	2.5–2.8	1
CP33	−37.86159	−71.0664	7	340.6	− 54.9	1.9	257	74.2	201.9	N	2.5–2.8	1
CP35	−37.81545	−71.1029	7	351.1	− 54.3	2.3	179	82.2	218.8	N	0.7–1.2	2
CP36	−37.81371	−71.1015	7	359.2	− 59.0	3.7	136	87.9	126.4	N	0.7–1.2	2
CP39	−37.88519	−71.0765	8	346.5	− 49.9	5.1	92	76.8	227.8	N	2.5–2.8	1
CP42	−37.87078	−70.9933	8	24.3	− 54.2	5.0	43	70.2	17.3	N	2.5–2.8	1
CP44	−37.84023	−71.0763	10	8.5	− 38.1	13.0	13	72.0	315.4	N	2.5–2.8	1
CP46	−37.93938	−71.0426	7	346.5	− 52.2	3.2	115	77.9	219.9	N	4.0–5.6	1
CP47	−37.90404	−71.0401	7	353.1	− 36.9	3.5	107	71.7	268.0	N	2.5–2.8	1
CP50	−37.97646	−70.7816	10	351.1	− 59.7	2.3	159	82.6	175.9	N	4.0–5.6	1
CP53	−37.84478	−71.1192	9	350.7	− 34.9	5.9	56	69.7	262.8	N	0.7–1.2	2
CP54	−37.84377	−71.1188	7	353.6	− 54.8	3.4	149	84.3	223.0	N	0.7–1.2	2
CP55	−37.88788	−71.1230	6	16.6	− 43.3	7.8	51	71.1	341.7	N	0.7–1.2	2
CP57	−37.84734	−71.1119	8	357.3	− 30.9	1.9	280	68.7	281.8	N	0.7–1.2	2
CP58	−37.85533	−71.1405	5	8.3	− 35.0	6.1	72	70.1	312.4	N	0.015–0.7	2
CP60	−37.84671	−71.1402	5	333.5	− 61.4	5.0	125	69.3	177.4	N	0–0.015	2
CP61	−37.82115	−71.0629	6	328.9	− 52.6	3.6	155	64.4	199.9	N	2.5–2.8	1
CP62	−37.82986	−70.9810	9	338.0	− 55.7	3.1	111	72.4	197.4	N	2.5–2.8	1
CP66	−37.84050	−70.9947	9	338.9	− 60.0	2.2	199	73.4	181.8	N	2.5–2.8	1
CP67	−37.84371	−70.9959	7	354.1	− 55.2	2.7	215	84.8	221.3	N	2.5–2.8	1
CP69	−37.78071	−71.0377	6	358.8	− 54.9	2.4	255	87.4	266.3	N	2.5–2.8	1
CP70	−37.78073	−71.0378	6	357.5	− 60.1	7.3	22	86.2	139.1	N	2.5–2.8	1
CP71	−37.79106	−71.0333	8	4.1	− 40.9	3.8	77	75.2	303.8	N	2.5–2.8	1
CP72	−37.81498	−71.0173	10	6.9	− 45.6	1.8	222	77.8	319.3	N	2.5–2.8	1
CP74	−37.86041	−71.0938	5	342.9	− 43.5	11.6	27	70.9	234.7	N	0.015–0.7	2
CP78	−37.85954	−71.1055	10	357.9	− 65.6	2.5	125	79.9	117.0	N	0.7–1.2	2
CP80	−37.85915	−71.1087	5	12.1	− 40.6	5.8	29	72.0	327.6	N	0.7–1.2	2
CP81	−37.84361	−71.1375	9	335.9	− 67.9	5.6	86	68.5	153.6	N	0–0.015	2
CP82	−37.84215	−71.1383	8	344.9	− 58.5	3.8	82	78.1	187.6	N	0–0.015	2
CP83	−37.84415	−71.1367	9	12.3	− 25.5	2.3	206	63.2	316.2	N	0–0.015	2
CP90	−37.87310	−71.0567	10	353.8	− 48.4	1.7	287	80.1	255.9	N	2.5–2.8	1

Notes: Slat and Slon are site latitude and longitude; *n* is number of specimens used in site mean calculations; Dec and Inc are declination and and inclination of site-mean direction; α_{95} is the radius of 95 per cent confidence cone; *k* is the Fisher (1953) precision parameter; Plat and Plon are corresponding VGP latitude and VGP longitude; P is the geomagnetic polarity: normal (N) and reversed (R); Age corresponds to the age range attributed to each site and Ref is the reference: 1. Linares *et al.* (1999); 2. Melnick *et al.* (2006).

5.1 Selection criteria

For further TAF and PSV analysis, our data set was separated into three groups according to values obtained for number of specimens per site (n) and Fisher precision parameter (k). This parameter is defined as a function of n and the length of the resultant vector (R): $k = (n - 1)/(n - R)$. The first data set considers all site-level data and other two subsets are differentiated by the k values used. The selection criteria employed for the three data sets are summarized below:

(i) No selection criteria and any k and n values were accepted.
(ii) Criteria employed by de Oliveira *et al.* (2021) for updated 0–10 Ma database, with $n \geq 5$ and $k \geq 50$.
(iii) Criteria proposed by Tauxe *et al.* (2003) that assumes $n \geq 5$ and $k \geq 100$.

These approaches will allow to examine whether the PSV and TAF estimates are influenced by incorporating of low-quality data, as reported by some studies (e.g. Biggin *et al.* 2008; Behar *et al.* 2019). It is worth mentioning that all palaeodirectional sites are assigned to the highest demagnetization code equal to 5 (DC = 5; McElhinny & McFadden 2000), this means that two demagnetization procedures (AFD and THD) were used with PCA.

5.2 Reversal test and transitional data

To evaluate whether two directions populations share a common mean direction, we used the reversal test proposed by McFadden & McElhinny (1990, hereafter MM90) for normal and reversed polarity data sets. This test provides a classification based on the angle θ between the two mean directions at the 95 per cent confidence level. A positive reversal test is labelled as A ($\theta \leq 5°$), B ($6° \leq \theta \leq 10°$), C ($10° \leq \theta \leq 20°$) and as indetermined for $\theta \geq 20°$. A similar test suggested by Watson (1983) also was applied to the normal and reversed data based on Monte Carlo simulation. MM90 and Watson's V tests were performed using the PmagPy package (https://earthref.org/PmagPy/).

In general, transitional data are not considered to investigate the palaeofield behaviour during stable polarity periods. As addressed by Johnson & McFadden (2015), the inclusion of transitional sites can yield an enhanced dispersion of directional data, and therefore can provide inadequate PSV and TAF estimates. We used the Vandamme (1994) method to identify transitional sites in our data sets.

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An optimum cut-off angle (A) from the palaeomagnetic pole, is determined as follows:

$$A(^\circ) = 1.8 S_B(^\circ) + 5^\circ. \quad (1)$$

An recursive method is applied to calculate A from the estimated VGP dispersion (S_B; see Section 5.3). VGP data with angular deviations greater than the A value are removed, then S_B is recalculated until there are no data with deviations greater than the value obtained from eq. (1).

5.3 Estimates of VGP dispersion and inclination anomaly

For the three data sets defined by the selection criteria, the between-site VGP dispersion (S_B) from N palaeodirectional sites was calculated to assess palaeosecular variation in the area studied, given by (e.g. Biggin *et al.* 2008):

$$S_B = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(\Delta_i^2 - \frac{{S_{wi}}^2}{n_i}\right)}, \quad (2)$$

where Δ_i is the angular distance between the ith VGP and the mean VGP (palaeopole); and n_i is the number of specimens per site and $S_{wi} = 81/K_i$ corresponds to the within-site VGP dispersion. K_i is determined as function of Fisher precision parameter (k_i) of site-mean directions and the site palaeolatitude (λ_i), following Cox (1970):

$$K_i = \frac{8k_i}{5 + 18\sin^2\lambda_i + 9\sin^4\lambda_i}. \quad (3)$$

To quantify directional deviations with respect to the GAD field, we calculated the inclination anomaly (ΔI) from the three data groups, which is determined by the difference between the Fisher (1953) mean inclination (I_{mean}) and the expected GAD inclination (I_{GAD}) using the following equation (Cox 1975):

$$\Delta I = I_{\mathrm{mean}} - I_{\mathrm{GAD}}, \quad (4)$$

where I_{GAD} is obtained from the mean latitude ($\bar{\lambda}$) for a given data set, defined as:

$$I_{GAD} = \tan^{-1}(2\tan\bar{\lambda}). \quad (5)$$

6 RESULTS

6.1 Rock magnetism

Thermomagnetic curves during heating cycles from 15 sites present two behaviour based on the Curie temperature (T_c) properties (Fig. 4a). The first is categorized as high T_c, with a sharp decrease in the magnetic susceptibility between 500 and 600 °C (represented by dotted curves in Fig. 4a), which indicates the presence of pure magnetite or Ti-poor titanomagnetite (Dunlop & Özdemir 1997; Evans & Heller 2003). The second reveals thermomagnetic curves (solid curves in Fig. 4a) with low transition temperatures of 180–330 °C (categorized as low T_c), which is typical of titanomagnetite with high Ti contents. Both patterns of magnetic mineralogy corroborate with the scanning electron microscopy results reported by Moncinhatto *et al.* (2020). Only samples from sites CP03 and CP21 have curves that are remarkably different from other samples. CP03 displays a higher increase of magnetic susceptibility during heating process, and has a single magnetic phase with $T_c \sim 574$ °C. Conversely, sample CP21 shows a secondary transition temperature of ∼330 °C, indicative of oxidized titanomagnetite or titanomaghemite (Dunlop & Özdemir 1997). See Table S2 (Supporting Information) for more details.

IRM acquisition curves were classified in two groups as discussed above. Curves associated with high T_c and low T_c present, respectively, a sharp and smooth change when saturation magnetization is reached (Fig. 4b). The saturation field values (<350 mT) suggest the presence of low coercivity minerals, such as magnetite, titanomagnetite and titanomaghemite (Dunlop & Özdemir 1997). Hysteresis loops are narrow for all samples and have low coercivity (B_c) values, ranging from 6 to 20 mT. Wasp-waisted magnetic behaviour is not observed, which could indicate a small contribution of paramagnetic minerals, as shown in Fig. 4(c) for hysteresis loop of CP04 site. FORC diagrams are typical of vortex state grains (Roberts *et al.* 2014, 2017; Lascu *et al.* 2018) with B_c values below 80 mT and a moderate spread of outer contours along the vertical (B_u) axis. Examples of FORC diagrams are shown in Figs 4(d)–(g).

6.2 Palaeomagnetic directional

ChRM directions were easily identified in most specimens for both AFD and THD procedures. Examples of demagnetization plots are reported in Fig. 5. Overall, ChRM components were isolated by AFD between 20–100 mT and THD from 200 °C until complete NRM demagnetization (∼600 °C), and defined from best-fitting lines of the demagnetization data directed towards the origin of the Zijderveld diagrams (Figs 5a–d). In some cases, ChRM directions were determined using the great circle method (McFadden & McElhinny 1988) for removal of secondary components, which are not identifiable by linear segments in the demagnetization plots. The statistical distribution of MAD and DANG estimates is shown in Fig. 6. A total of 678 specimens were thermally and AF demagnetized, of which 515 specimens were determined the ChRM direction with best-fitting lines. Among these, 433 specimens (∼ 84 per cent) have MAD and DANG values lower than 5°. For the remaining 163 specimens, ChRM directions were obtained from best-fitting great circles of demagnetization data, and only 50 per cent of the specimens have values of MAD $\leq 5^\circ$.

Eight sites (CP52, CP56, CP63–64, CP68, CP76–77, CP79) out of the 50 sampled palaeomagnetic sites do not yield results from demagnetization diagrams analysis due to high scattered palaeodirections, and consequently, site-level data have no statistical significance. The observed dispersion can be caused by some geological characteristics arising from devolatization (presence of vesicles) and brecciation (resulted from interactions between lava and ice). The 42 site-mean directions along with corresponding VGPs are summarized in Table 1. There are 36 sites with normal polarity and 6 sites have reversed polarity. In the latter, all data are assigned to the Cola de Zorro Formation. Great circle fits were used only 15 per cent of the specimens to determine site-mean directions. Equal-area stereographic projections of site-mean directions for the three data sets defined by the selection criterion described in Section 5.1 are shown in Figs 7(a)–(c). The VGP distributions are represented in the polar projection maps in Figs 7(d)–(f). It is noteworthy that all site directional data were calculated with at least 5 or more specimens (see Table 1). Thus, we use only the Fisher precision parameter, k, as a cut-off value for filtered data sets. The normal and reversed site-mean directions for the three groups pass a bootstrap reversal test (Tauxe 2010) at the 95 per cent confidence level (Fig. S1, Supporting Information), and Watson's V-test with C classification (Fig. 8) of MM90. This enables the mean direction and palaeopole to be determined by combining normal data and

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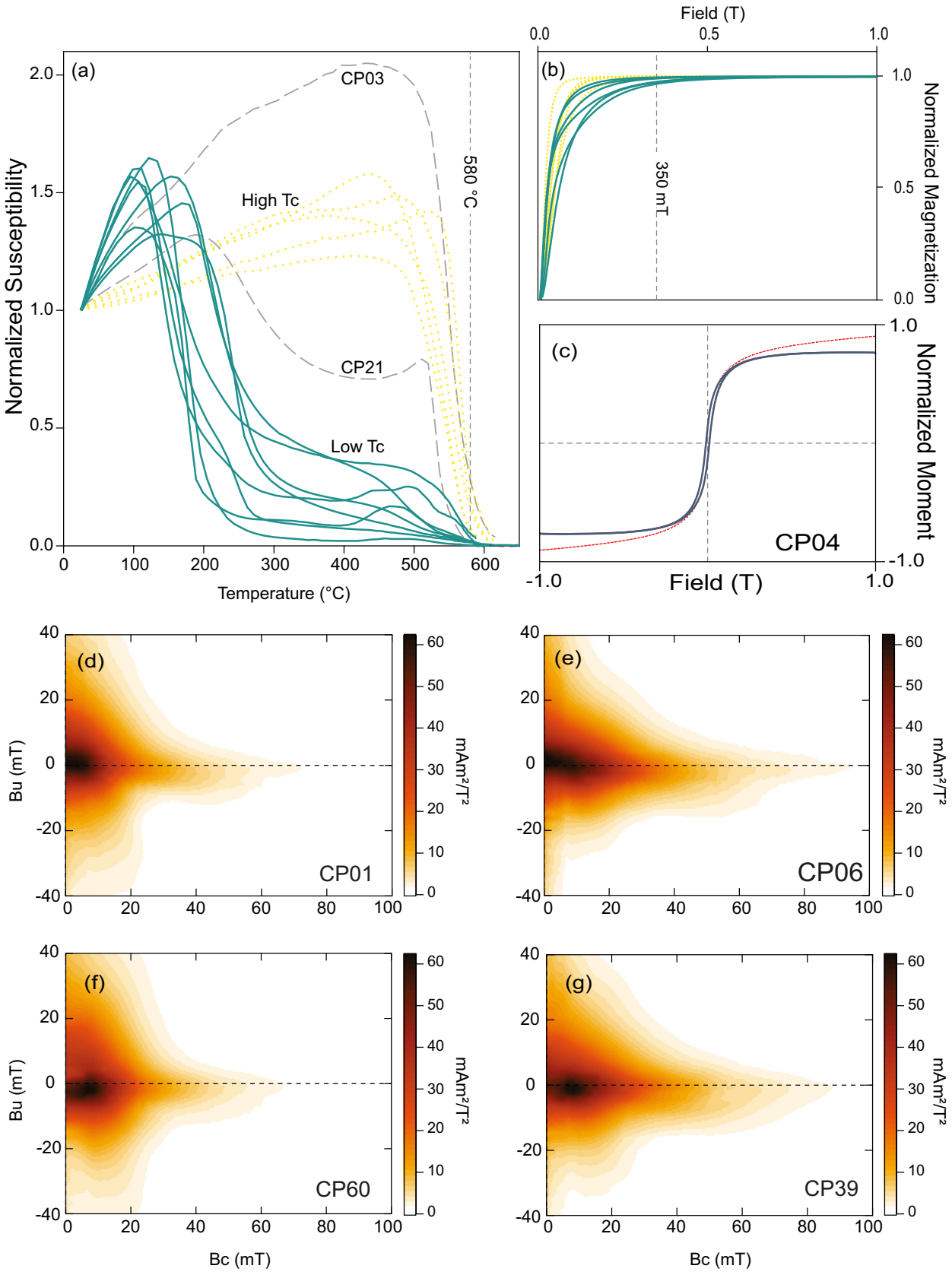


Figure 4. Magnetic mineralogy measurements. (a) Thermomagnetic curves grouped according to the Curie temperature (T_c) properties: high T_c (dotted curve), low T_c (solid curves), dashed curves (sites CP03 and CP21) have different characteristics than the others groups. (b) IRM acquisition curves grouped as in (a). (c) Hysteresis cycles with (solid curve) and without (dotted curve) paramagnetic correction. (d)–(g) Conventional FORC diagrams. IRM = isothermal remanent magnetization.

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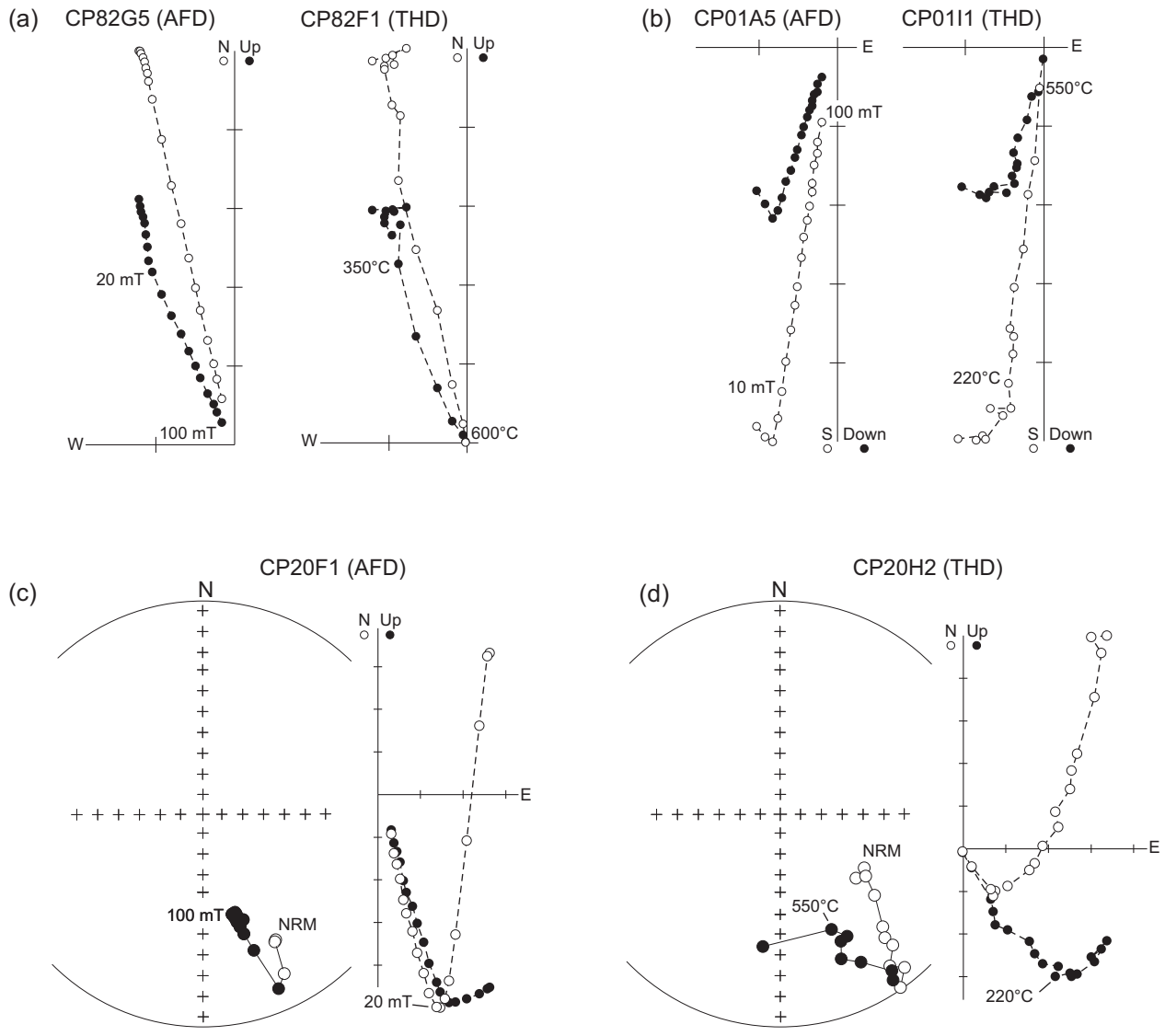


Figure 5. Examples of demagnetization diagrams. Zijderveld projections obtained from AFD and THD for (a) normal and (b) reversed polarity directions. Zijderveld and stereographic projections of a reverse specimen during (c) AFD and (d) THD. ChRM = characteristic remanent magnetization, AFD = alternating field demagnetization and THD = thermal demagnetization.

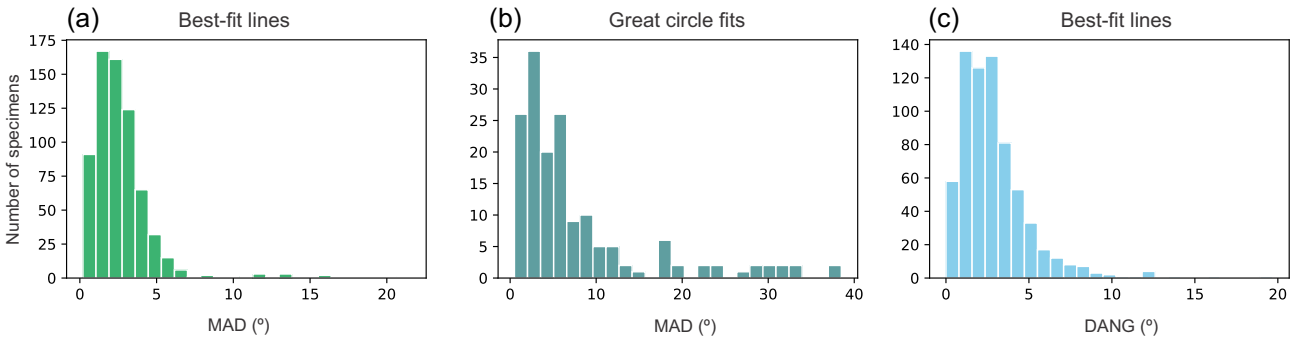


Figure 6. Distributions of MAD estimates from (a) best-fitting lines and (b) great circle fits, and for (c) DANG estimates from best-fitting lines that yielded ChRM directions. MAD = maximum angular deviation, DANG = deviation angle and ChRM = characteristic remanent magnetization.

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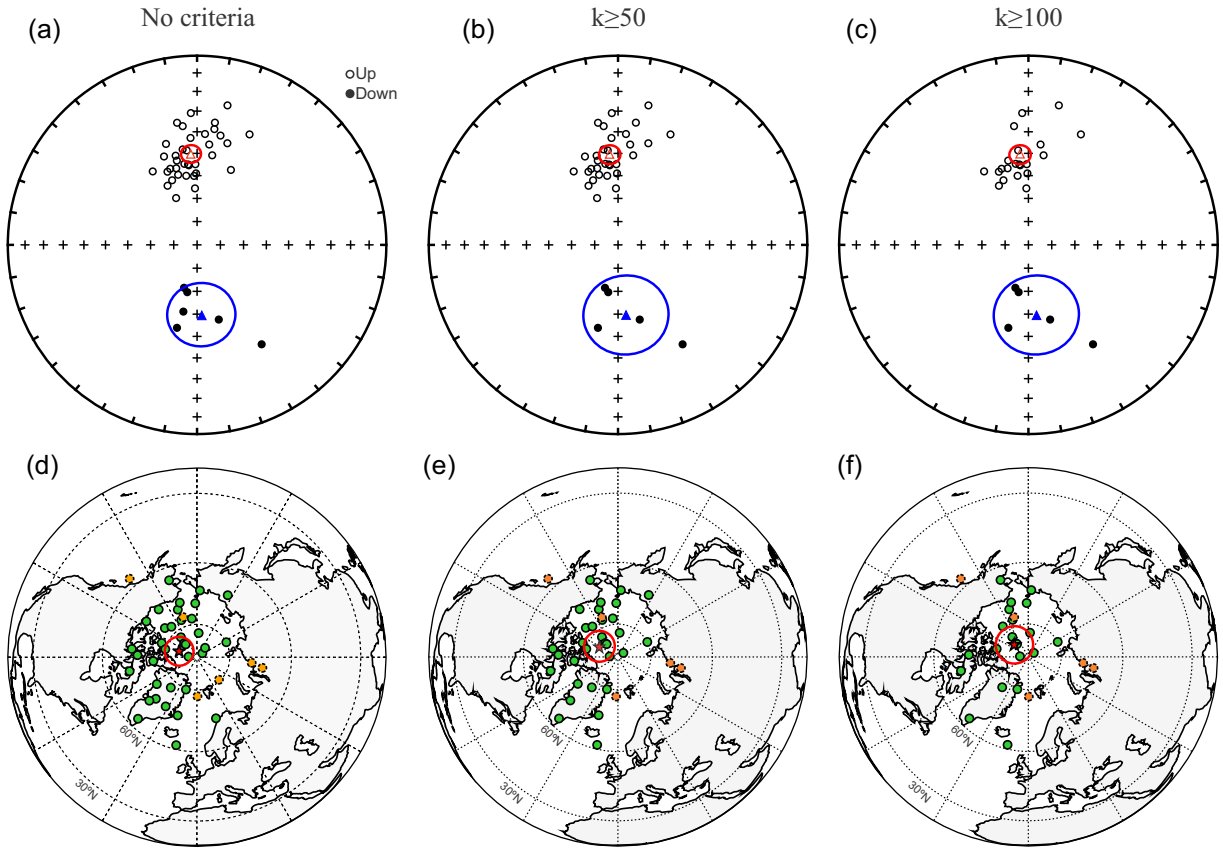


Figure 7. Stereographic projections for normal (open circles) and reversed (filled circles) site-mean directions from three data sets with (a) no selection criteria, (b) $k \geq 50$ and (c) $k \geq 100$. Open (filled) triangles indicate the normal (reversed) mean directions with their 95 per cent confidence circles. (d)–(f) Polar stereographic maps of VGP positions for each data set. Solid (dashed) circles refer to normal (reversed) polarity VGPs. The stars represent the palaeomagnetic poles with 95 per cent confidence circles. VGP = virtual geomagnetic pole.

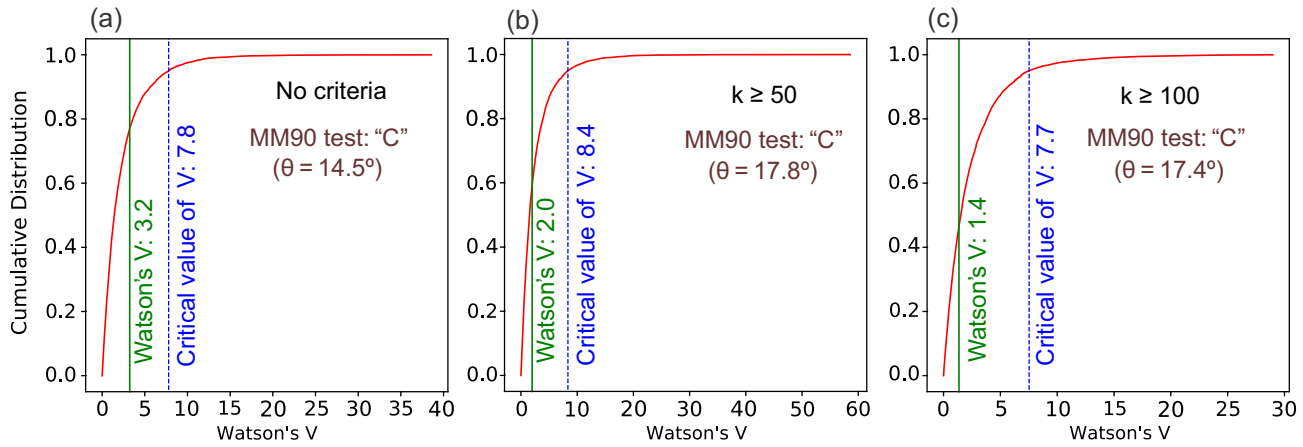


Figure 8. Watson (1983) V-test and MM90 (McFadden & McElhinny 1990) test results for the three data sets with (a) no selection criteria, (b) $k \geq 50$ and (c) $k \geq 100$. The normal and reversed directions share a common mean, with a C classification (see Section 5.2 for details).

antipodes of the reversed data (Table 2). The three data sets cover the same age range of 0.015–5.6 Ma.

For the data set without selection criteria ($N = 42$ sites), the mean direction from normal (reversed) sites is $D = 356.2°, I = -50.1°, \alpha_{95} = 4.0°$ and $N = 36$ ($D = 176.5°, I = 59.5°, \alpha_{95} = 14.1°$ and $N = 6$). The overall mean direction ($D = 356.2°, I = -51.5°$ and $\alpha_{95} = 3.9°$) does not coincide at the 95 per cent confidence level with expected GAD direction ($I_{GAD} = -57.3°$). Similarly, the mean VGP location (Plat = 84.3°, Plon = 251.4° and $A_{95} = 4.4°$) is not statistically compatible with the geographic north pole (Fig. 7d). Applying the selection criteria $k \geq 50$, the mean direction for normal (reversed) polarity sites is $D = 354.7°, I = -50.4°, \alpha_{95} = 4.2°$ and $N = 31$ ($D = 173.5°, I = 59.0°, \alpha_{95} = 17.7°$ and $N = 5$). When using $N = 36$ sites, the mean direction ($D = 354.6°, I = -51.6°$ and $\alpha_{95} = 3.9°$) and mean VGP (Plat = 83.5°, Plon = 240.1° and $A_{95} = 4.7°$) do not overlap with the predicted GAD direction and the geographic north

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Table 2. Palaeomagnetic mean directions for the three data sets.

Data set	N	Slat (°N)	Slon (°E)	D (°)	I (°)	k	α_{95} (°)	Plat (°N)	Plon (°E)	K	A_{95} (°)	S_B (°)	S_{Bl} (°)	S_B^u (°)	ΔI (°)	ΔI_{lo} (°)	ΔI^{up} (°)
No selection criteria																	
All data	42	−37.86	−71.03	356.2	−51.5	32.8	3.9	84.3	251.4	26.4	4.4	15.3	13.0	17.4	5.8	3.1	9.7
Normal	36	−37.85	−71.06	356.1	−50.1	36.0	4.0	83.0	255.5	30.2	4.4	14.3	11.8	16.6	7.1	3.2	11.0
Reverse	6	−37.93	−70.85	176.5	59.5	23.4	14.1	−86.1	303.4	15.8	17.4	19.8	11.9	27.9	2.2	−1.7	6.1
$k \geq 50$																	
All data	36	−37.86	−71.03	354.6	−51.6	32.7	4.2	83.5	240.1	27.0	4.7	15.4	12.9	17.9	5.7	1.5	9.9
Normal	31	−37.85	−71.06	354.7	−50.4	36.6	4.3	82.4	245.8	31.8	4.7	14.1	11.3	16.4	6.8	2.6	11.0
Reverse	5	−37.93	−70.86	173.5	59.0	19.7	17.7	−85.2	327.9	13.3	21.7	22.1	15.0	30.1	1.7	−2.5	5.9
$k \geq 100$																	
All data	26	−37.87	−71.01	354.4	−53.2	31.5	5.1	84.4	229.1	25.7	5.7	15.8	11.8	18.9	4.1	−1.0	9.2
Normal	21	−37.85	−71.05	354.5	−51.8	35.7	5.4	83.2	238.3	31.6	5.7	14.2	10.5	17.9	5.5	0.4	10.6
Reverse	5	−37.93	−70.86	173.5	59.0	19.7	17.7	−85.2	327.9	13.3	21.7	22.1	15.0	30.1	1.7	−3.4	6.8

Notes: N is the number of sites; Slat and Slon are the mean site latitude and longitude; D and I denote the mean declination and inclination; k and α_{95} are corresponding 95 per cent confidence cone and Fisher (1953) precision parameter of the mean direction; Plat and Plon are the latitude and longitude of the palaeomagnetic pole; K and A_{95} are the 95 per cent confidence cone and Fisher (1953) precision parameter of the palaeomagnetic pole; S_B is the between-site VGP dispersion with its lower (S_{Bl}) and upper (S_B^u) 95 per cent confidence bounds; ΔI is the inclination anomaly with lower (ΔI_{lo}) and upper (ΔI^{up}) 95 per cent confidence bounds.

pole within the 95 per cent confidence interval, respectively (Table 2 and Fig. 7e). Finally, the data set filtered with $k \geq 100$ consists of 26 sites. Among these, the mean direction for normal sites ($N = 21$) is $D = 354.5°$, $I = -51.8°$ and $\alpha_{95} = 5.4°$ and for reversed sites ($N = 5$) is $D = 173.5°$, $I = 59.0°$ and $\alpha_{95} = 17.7°$. The mean direction ($D = 354.4°$, $I = -53.2°$ and $\alpha_{95} = 5.1°$) falls close to the expected GAD direction, whereas the mean VGP (Plat = 84.4°, Plon = 229.1° and $A_{95} = 5.7°$) is statistically indistinguishable at the 95 per cent confidence level with the geographic north pole (Fig. 7f).

No transtional sites are identified for the three data sets when applying the Vandamme (1994) criterion. S_B and ΔI estimates with upper (S_B^u and ΔI^{up}) and lower (S_{Bl} and ΔI_{lo}) 95 per cent confidence bounds for each data group are presented in Table 2. Our data set with no selection criteria yields values of $S_B = 15.3^{17.4^\circ}_{13.0}$ and $\Delta I = 5.8^{9.7^\circ}_{3.1}$ that are not statistically different within the 95 per cent confidence limits compared to the estimates for data sets filtered with $k \geq 50$ ($S_B = 15.4^{17.9^\circ}_{12.9}$ and $\Delta I = 5.7^{9.9^\circ}_{1.5}$) and $k \geq 100$ ($S_B = 15.8^{18.9^\circ}_{11.8}$ and $\Delta I = 4.1^{9.2^\circ}_{-1.0}$).

7 DISCUSSION

7.1 Magnetostratigraphy of the Caviahue–Copahue Volcanic Field

Sampled sites cover an age range of approximately 5.6 Ma (between 0.015–5.6 Ma), and comprise the last four polarity chrons (Brunhes, Gauss, Matuyama and Gilbert) during the Late Miocene and the Holocene. As shown in Fig. 9, the magnetic polarity of the site-mean directions are in accordance with the geomagnetic polarity time scale 2020 (GTPS2020) of Ogg (2020), taking into account the age interval attributed to each site, supported by geochronological studies listed in Table 1. Most of palaeodirectional sites have normal polarity and are associated with the lava flows from Mellizas Sequence and Copahue Volcano (Fig. 9). Reversed polarity directions are recorded only for CP01-05 and CP20 sites and represent the Cola de Zorra formation.

7.2 Previous results from South America

In their recent compilation of palaeomagnetic data from volcanic sites, Cromwell *et al.* (2018) and de Oliveira *et al.* (2021) reinforced the need of globally distributed high-quality data, in order to better solve TAF structures in terms of zonal and non-zonal contributions. This lack of reliable data is mainly observed in the Southern Hemisphere, where TAF and PSV approaches are supported by an uneven palaeodirection data distribution when compared to the Northern Hemisphere. The current 0–10 Ma database (de Oliveira *et al.* 2021) comprises 2543 palaeodirectional data from lava flows, of which 1447 (65 per cent) are from the Northern Hemisphere and only 788 (35 per cent) are from the Southern Hemisphere. In South America, palaeodirectional contributions began in 1994 with the work of Cañón-Tapia *et al.* (1994) followed by some studies from the 2000s (Leonhardt *et al.* 2003; Mejia *et al.* 2004; Brown *et al.* 2004, 2021; Opdyke *et al.* 2006; Quidelleur *et al.* 2009; Roperch *et al.* 2015; Sánchez-Duque *et al.* 2016; de Oliveira *et al.* 2022). A summary of palaeomagnetic results from the current 0–10 Ma database (de Oliveira *et al.* 2021) and new PSV studies is reported in Table 3 and plotted in Fig. 10.

As can be observed in Figs 10(a) and (b), data from South America (SA) covering latitudes between 5–20°S are scarce. Therefore, future TAF and PSV studies should concentrate on acquiring new palaeomagnetic records in this latitude band. We have five studies at latitudes 23–51°S: three are from Argentina (Brown *et al.* 2004; Mejia *et al.* 2004; Quidelleur *et al.* 2009) and two from Chile (Roperch *et al.* 2015; Brown *et al.* 2021). Inclination anomaly estimates these studies are consistent at the 95 per cent confidence level with a GAD field ($\Delta I = 0°$, Fig. 10b and Table 3). All results are not also statistically different from predictions of recent palaeomagnetic models: 0–10 Ma TAF (de Oliveira *et al.* 2021), BCE19 (Brandt *et al.* 2020) and BB18-family (Bono *et al.* 2020) models. For low latitudes in SA region, there are four studies from Brazil (Leonhardt *et al.* 2003), Ecuador (Opdyke *et al.* 2006) and Colombia (Sánchez-Duque *et al.* 2016; de Oliveira *et al.* 2022) that yielded negative ΔI values with significant deviations from a GAD model, expect the study of de Oliveira *et al.* (2022). In addition, note that inclination anomalies are statistically compatible with the 0–10 Ma TAF and BB18.Z3 models, considering the 95 per cent confidence intervals (Fig. 10b). Although SA inclination anomalies present local persistent departures from the predictions discussed above, regional data indicate consistency with the 0–10 Ma data compilation including our results (Table 2).

Regarding the PSV estimates (Fig. 10a), two studies at intermediate latitudes from Argentina (Mejia *et al.* 2004; Quidelleur *et al.* 2009) are in agreement with VGP dispersion curves of the BCE19

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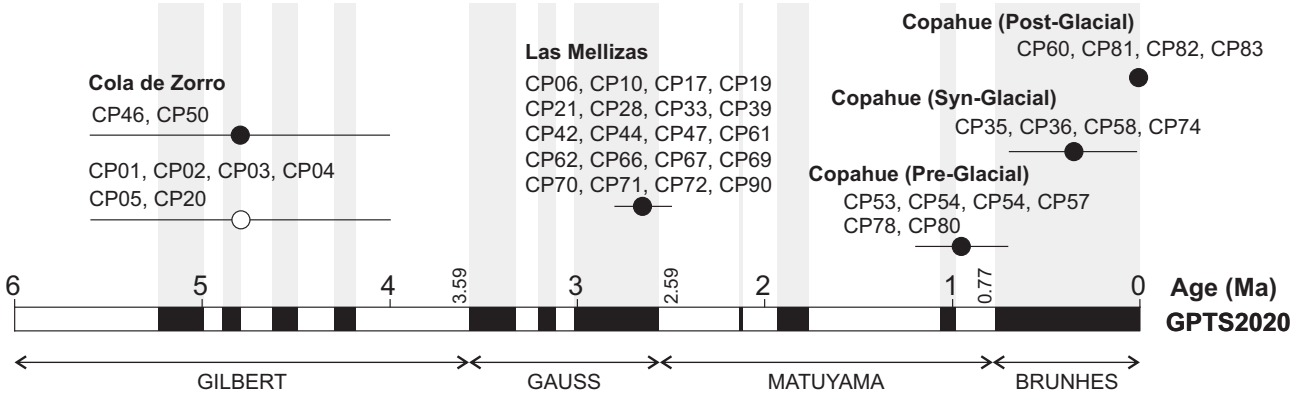


Figure 9. Magnetic polarity of the sampled sites compared to the geomagnetic polarity time scale 2020 (GPTS2020; Ogg 2020). The age interval of each site (see Table 1) is represented by a horizontal line. Normal (reversed) polarity sites are indicated by filled (open) circles.

Table 3. Mean palaeodirections from South American studies. Reference refers to the palaeomagnetic study conducted in South America (see column location).

Nos	Location	Age (Ma)	Slat (°N)	Slon (°E)	N	D (°)	I (°)	α_{95} (°)	Plat (°N)	Plon (°E)	A_{95} (°)	S^{u}_{Bl} (°)	ΔI^{up}_{lo} (°)	Reference
1	Argentina	0.165–8.67	− 51.39	289.25	36	357.4	− 68.1	3.6	88.2	178.3	5.6	$18.6^{20.7}_{15.5}$	$0.2^{2.7}_{-2.2}$	Mejia *et al.* (2004)
2	Argentina	0.034–7.86	− 47.07	288.78	26	5.3	− 63.9	5.6	86.2	36.1	7.7	$21.3^{23.8}_{17.3}$	$1.1^{6.4}_{-3.8}$	Brown *et al.* (2004)
3	Argentina	0.007–1.72	− 36.14	290.85	30	357.3	− 54.4	4.7	88.0	197.1	4.9	$14.7^{17.7}_{11.1}$	$1.2^{6.1}_{-2.6}$	Quidelleur *et al.* (2009)
4	Chile	0.00016–0.0051	− 38.90	288.24	18	5.2	− 55.0	4.4	85.6	342.9	4.9	$11.3^{12.9}_{8.7}$	$3.2^{7.5}_{-0.2}$	Roperch *et al.* (2015)
5	Chile	0.1–4.2	− 23.30	292.20	39	358.9	− 37.6	5.8	88.3	240.5	5.8	$18.5^{21.9}_{16.0}$	$-2.7^{3.1}_{-8.5}$	Brown *et al.* (2021)
6	Brazil	2.55–10.0	− 4.32	327.74	36	358.9	− 15.4	4.8	86.1	165.8	3.5	$11.7^{13.5}_{9.7}$	$-6.8^{-1.9}_{-10.8}$	Leonhardt *et al.* (2003)
7	Ecuador	0.0176–2.71	− 0.48	281.76	45	0.6	− 6.6	4.0	87.0	89.5	3.3	$12.5^{14.1}_{10.5}$	$-5.6^{-2.5}_{-8.6}$	Opdyke *et al.* (2006)
8	Colombia	0–2.02	1.27	282.76	30	351.2	− 3.4	6.2	80.7	173.1	5.2	$15.2^{17.6}_{12.0}$	$-5.9^{0.3}_{-12.1}$	de Oliveira *et al.* (2022)
9	Colombia	0–2.65	4.90	284.64	42	3.9	3.8	4.3	85.1	51.7	2.9	$10.5^{11.9}_{8.9}$	$-5.9^{-2.2}_{-10.1}$	Sánchez-Duque *et al.* (2016)

Note: See Table 2 for a complete description of columns Age to ΔI.

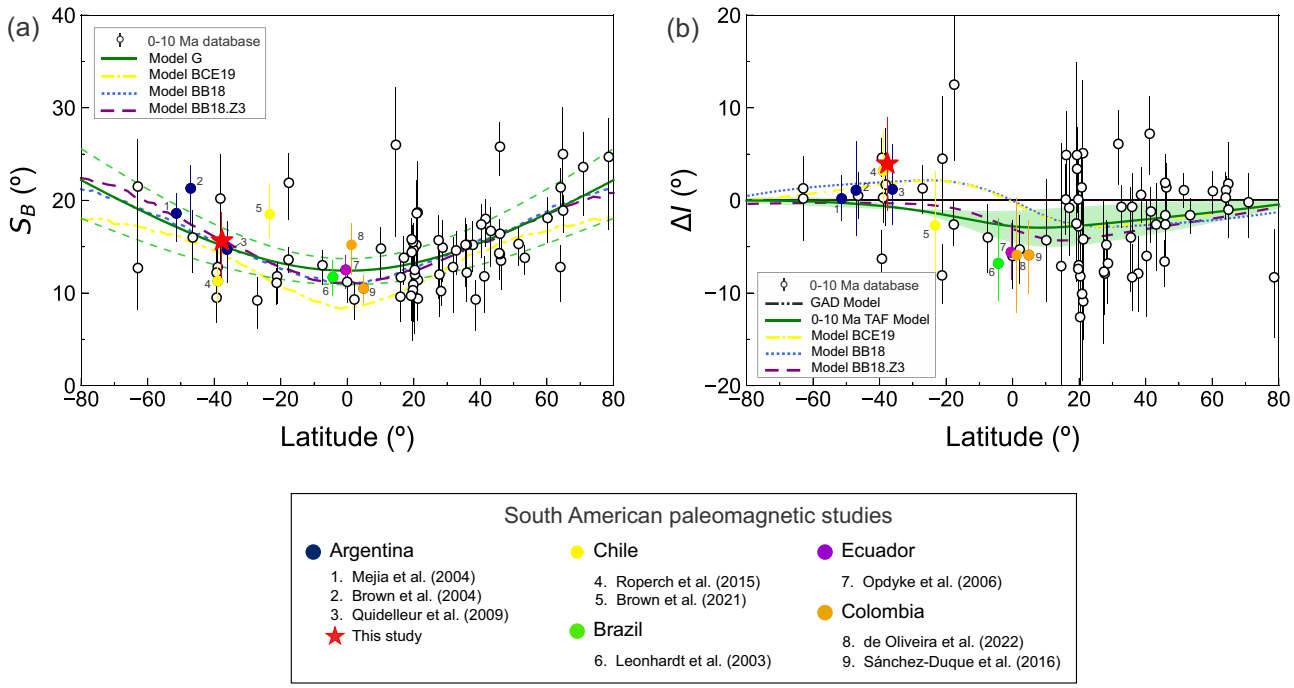


Figure 10. Latitudinal distribution of (a) VGP angular dispersion (S_B) and (b) Inclination anomaly (ΔI) from updated 0–10 Ma database (de Oliveira *et al.* 2021) filtered with Vandamme (1994) criterion. The stars represent the S_B and ΔI estimates of this study. Numbered circles indicate data from South American studies, as reported in Table 3.

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and BB18 models, and revised Model G (McFadden & McElhinny 1988) suggested by de Oliveira *et al.* (2021). A low dispersion ($S_B = 11.7^{13.5^\circ}_{9.7}$) is found for the data set of Roperch *et al.* (2015), possibly due to the very short time span (historic) represented by the sampled flows. The studies by Brown *et al.* (2004) and Brown *et al.* (2021) have high VGP dispersion ($S_B = 21.3^{23.8^\circ}_{17.3}$ and $18.5^{21.9^\circ}_{16.0}$, respectively). These values are only statistically compatible with Model G within the 95 per cent confidence interval (dashed curve; Fig. 10a), the exception is the high PSV from Chile at 23°S (Brown *et al.* 2021). Interestingly, a pronounced dispersion is detected at a similar latitude from Saint Helena island ($S_B = 21.9^{25.1^\circ}_{17.9}$) at 18°S (Engbers *et al.* 2020) for the 8–10 Ma period, which suggests a long-term anomalous geomagnetic field behaviour in the Southern Hemisphere, as pointed out by some authors (e.g. Panovska *et al.* 2019; Engbers *et al.* 2020; Brown *et al.* 2021; de Oliveira *et al.* 2021). The studies from Brazil (Leonhardt *et al.* 2003), Ecuador (Opdyke *et al.* 2006) and Colombia (Sánchez-Duque *et al.* 2016) conform with BB18 models and Model G. The exception is BCE19 model that provides lower VGP dispersion estimates relative to other PSV models.

7.3 Time-averaged field and palaeosecular variation

We note that inclination anomalies this study have positive values for the three data sets, but they are not statistically different from each other (see Table 2). Using the database filtered with Tauxe *et al.* (2003) criterion, we obtain $\Delta I = 4.1^{9.2^\circ}_{-1.0}$ (represented by star) that is statistically compatible at the 95 per cent confidence level with the predictions of 0–10 Ma TAF model and two Giant Gaussian Process models (BCE19 and BB18 models), as shown in Fig. 10(b). Our results are suggestive of the presence of non-dipole field contributions, as reported by these models. In particular, for BCE19 (dotted–dashed curve) and BB18 (dotted curve) models that assume an axial octupole contribution of ~ 3 per cent of the axial dipole term.

VGP dispersion calculated following the Tauxe *et al.* (2003) criteria with value of $S_B = 15.8^{18.9^\circ}_{11.8}$, is statistically indistinguishable at the 95 per cent confidence level compared to the data sets without selection criteria and with $k \geq 50$ (see Table 2). By comparing the VGP dispersion curves from BB18 (Bono *et al.* 2020) models and Model G, our S_B estimate fits these PSV models within the 95 per cent confidence limits (Fig. 10a). Therefore, it suggest that we adequately sampled the palaeosecular variation during the last five million years.

8 CONCLUSIONS

We provide new high-quality palaeomagnetic data from the CCVC covering the past 5 Myr. The data set consists of 42 palaeodirectional sites from lava flows. The magnetic mineralogy measurements suggest the dominance of titanomagnetite with variable Ti contents with domain structures characteristic of vortex state grains. We obtain 26 palaeodirectional data after applying the strict selection criteria of $k \geq 100$, which have a mean direction ($D = 354.4^\circ$, $I = -53.2^\circ$ and $\alpha_{95} = 5.1^\circ$) very close to the GAD field direction ($D_{\mathrm{GAD}} = 0^\circ$ and $I_{\mathrm{GAD}} = -57.3^\circ$). The palaeomagmetic pole (Plat $= 84.4^\circ$, Plon $= 229.1^\circ$, $A_{95} = 5.7^\circ$) is statistically indistinguishable at the 95 per cent confidence level with the geographic north pole. The inclination anomaly ($\Delta I = 4.1^{9.2^\circ}_{-1.0}$) is agreement with predictions of 0–10 Ma TAF model, BCE19 and BB18 models at the 95 per cent confidence level. Our results support non-GAD field structures described by these models for the 0–10 Ma period. VGP dispersion ($S_B = 15.8^{18.9^\circ}_{11.8}$) is in accordance with global PSV models and South American studies, which suggests that PSV models are able to describe the palaeomagnetic observations for the latitude range of the studied area.

This study offers a new reliable data set for South America, where palaeomagnetic data coverage is particularly poor for the past 10 Myr. We reinforce the need for an upgraded 0–10 Ma data set that meet modern laboratory protocols, not only in the South American region, but for the entire Southern Hemisphere. Improvements in the geographic distribution of directional and intensity data, accompanied by reliable geochronologic studies are essential to develop a new generation of global TAF and PSV models. Moreover, a continuous latitudinal coverage of palaeomagnetic data may help to understand the long-period PSV and TAF asymmetries and to properly discern whether it is an intrinsic feature of the geomagnetic field, or is an artefact due to the low distribution and quality of the available palaeomagnetic data, especially in the Southern Hemisphere.

ACKNOWLEDGMENTS

The authors acknowledge the comments and suggestions of Harald Böhnel and Laurie Brown, and the editorial work of Eduard Petrovsky. This work was supported by grants of Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq; 150796/2022-6, 165161/2018-3, 425728/2018-8, 312737/2020-3). This work was partially funded by grant 2022/03086-2 to GAH, São Paulo Research Foundation (FAPESP). This study has been (partially) performed at USPMag Lab at Instituto de Astronomia, Geofísica e Ciências Atmosféricas (IAG) at Universidade de São Paulo (USP) funded by CAPES/FAPESP/CNPq.

DATA AVAILABILITY STATEMENT

The authors confirm that the data supporting the findings of this study are available within the paper and its Supporting Information.

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SUPPORTING INFORMATION

Supplementary data are available at *GJI* online.

Figure S1. Bootstrap reversal tests represented by cumulative distributions of Cartesian components for three data sets defined by the selection criteria: (a) No selection criteria, (b) $k \geq 50$ and (c) $k \geq 100$. Curved solid (dashed) lines refer to the normal (reversed) polarity directions. Vertical straight lines are the 95 per cent confidence limits.

Table S1. K-Ar data from geochronological studies carried out at the CCVC. Lithologies: andesite (AN); basaltic-andesite (BA); dacite (DA); daciticignimbrite (DI); traquiandesite (TA). Methods: whole-rock (WR).

Table S2. Rock magnetism data from thermomagnetic susceptibility ($\chi(T)$) curves.

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